



**Year 11 Mathematics Advanced
Assessment Task 2
Term 3 Week 3, 2023**

Setter: EKA

Name: _____

Teacher: _____

- Answer Questions in the space provided.
- Answer multiple choice questions on the question paper by circling the correct alternative.
- Show all relevant mathematical reasoning and/or calculations.
- Use blue or black PEN only.
- NESA approved calculators are permitted in this task.
- Marks may be deducted for insufficient or illegible work.
- Extra writing space is provided at the back of this booklet.
If you use this space, clearly indicate which question you are answering.
- Total possible marks: **32**
- Time allowed: **45 minutes plus 2 minutes Reading Time**

Wednesday, 2nd August 2023

Q1	Q2	Q3	Q4	Q5	Q6	Q7	TOTAL
/6	/6	/4	/4	/4	/4	/4	/32

Question 1 (6 marks)

Marks

- (a) Circle your correct alternative.

1

The graph of the function $f(x) = 18$ intersects with the graph of $g(x) = 2^x$ where:

(A) $x = \log_2 18$

(B) $x = \frac{1}{18}$

(C) $x = 9$

(D) $x = \log_{18} 2$

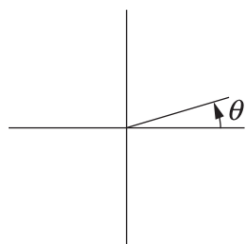
- (b) Circle your correct alternative.

1

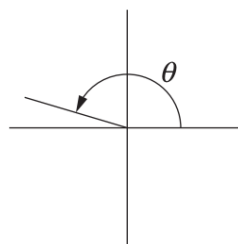
For the angle θ , $\sin \theta = \frac{7}{25}$ and $\cos \theta = -\frac{24}{25}$.

Which diagram best shows the angle θ ?

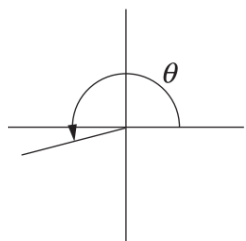
(A)



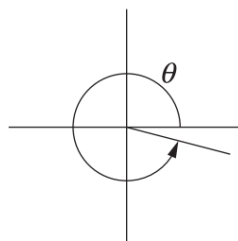
(B)



(C)



(D)



- (c) Solve $\cos \theta = 0.63$ for $-180^\circ \leq \theta \leq 0^\circ$, giving your answer to the nearest degree.

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Question 1 Continued

- (d) Find the derivative of $y = \frac{3x^4 - 1}{x}$.

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- (e) Given that $f(x) = x^2 + 1$ and $g(x) = \sqrt{x - 3}$. Find $y = f(g(x))$ and state its domain.

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End of Question 1

Question 2 (6 marks)

Marks

- (a) Circle your correct alternative.

1

What is the exact value of $\cos 135^\circ + \operatorname{cosec} 60^\circ$?

(A) $\frac{2\sqrt{2} - \sqrt{3}}{\sqrt{6}}$

(B) $\frac{2\sqrt{2} - 1}{\sqrt{2}}$

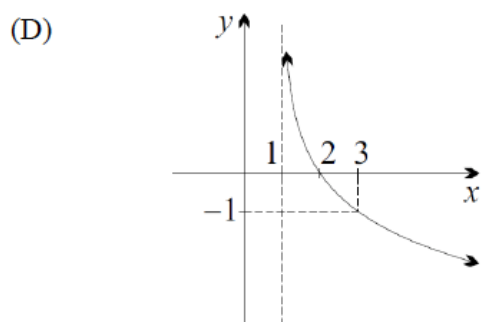
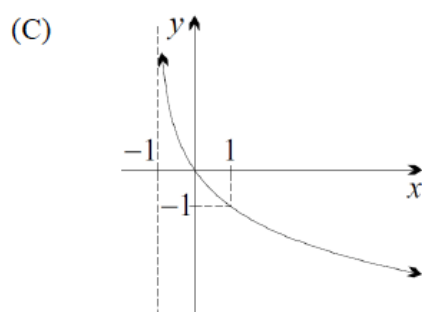
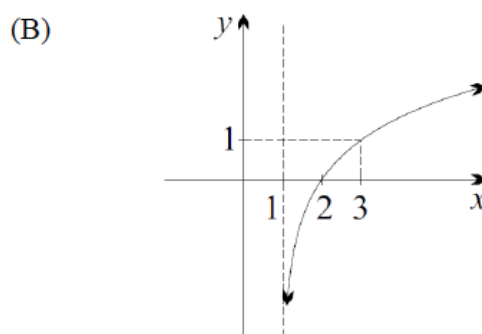
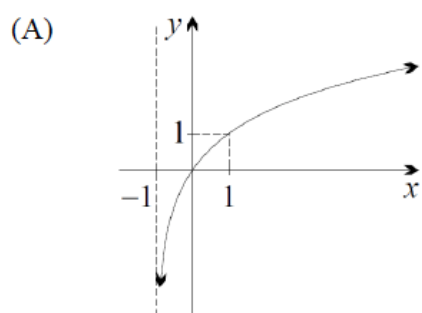
(C) $\frac{2\sqrt{2} + \sqrt{3}}{\sqrt{6}}$

(D) $\frac{2\sqrt{2} + 1}{\sqrt{2}}$

- (b) Circle your correct alternative.

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Which of the following is the graph of $y = -\log_2(x+1)$?



Question 2 Continued

- (c) Solve the equation $|2x - 3| = 6$.

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- (d) Differentiate with respect to x . Leave your answer in fully factorised form.

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$$y = \frac{2x^2}{2 - x}$$

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End of Question 2

Question 3 (4 marks)**Marks**

(a) Solve $\sin\left(\frac{x}{2}\right) = \frac{1}{2}$ for $0^\circ \leq x \leq 360^\circ$.

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(b) Show that $\cot A + \frac{\sin A}{1 + \cos A} = \operatorname{cosec} A$.

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End of Question 3

Question 4 (4 marks)

Marks

- (a) i) Show that the derivative of $f(x) = x^2 + 3x + 5$ is $f'(x) = 2x + 3$ by **first principles**.

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- ii) Hence, find the equation of the normal to the curve $f(x) = x^2 + 3x + 5$ at the point $(1, 9)$.

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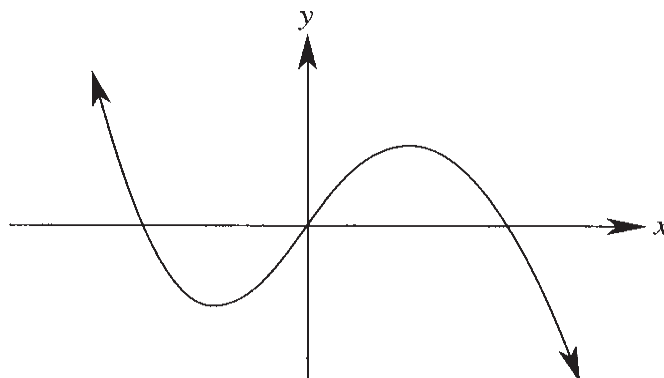
End of Question 4

Question 5 (4 marks)

Marks

- (a) The diagram below shows a sketch of the function $y = f(x)$.
On the same diagram sketch the gradient function $y = f'(x)$.

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- (b) Show that, $\frac{d}{dx} [3x^2(4x - 1)^3] = 6x(10x - 1)(4x - 1)^2$.

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End of Question 5

Question 6 (4 marks)**Marks**

- (a) Differentiate with respect to x .

$$y = \sqrt{x^2 - 2x}$$

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- (b) Solve $(\log_2 x)^2 - 8 = \log_2 x^2$.

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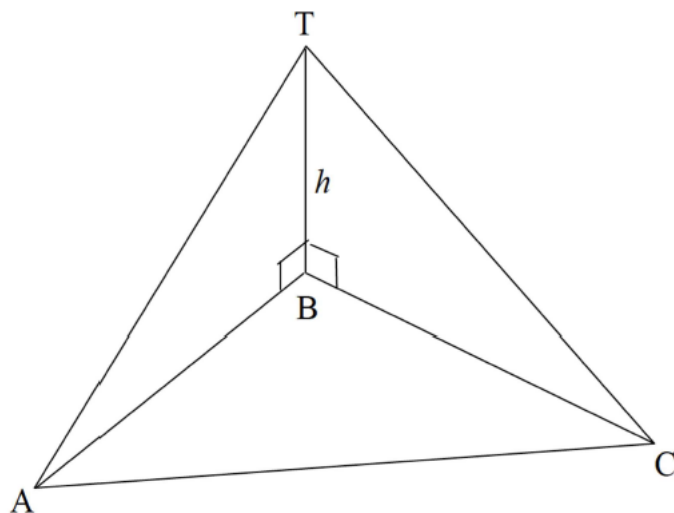
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End of Question 6

The angle of elevation of a vertical tower BT of height h metres, at a point A due west of it, is 14° . From the point C , the bearing of the tower is 343° and the angle of elevation is 11° . The points A and C are 1000m apart and on the same level as the base B of the tower.



- (a) By drawing a diagram showing the bearings, explain why $\angle ABC = 107^\circ$.

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- (b) Show that $AB = h \tan 76^\circ$ and find a similar expression for CB .

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(c) Hence, find the height of the tower, correct to the nearest metre.

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End of Question 7

End of Assessment Task

Extra writing space

If you are using this space indicate clearly which question(s) you are answering & write ‘see extra writing space’ on the original question space provided for that question.

[illegible]



**Year 11 Mathematics Advanced
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Question 1 (6 marks)

Marks

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The graph of the function $f(x) = 18$ intersects with the graph of $g(x) = 2^x$ where:

(A)

$x = \log_2 18$

(B)

$x = \frac{1}{18}$

(C)

$x = 9$

(D)

$x = \log_{18} 2$

$$2^x = 18$$

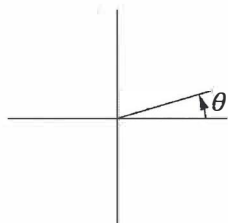
$$x = \log_2 18$$

- (b) Circle your correct alternative.

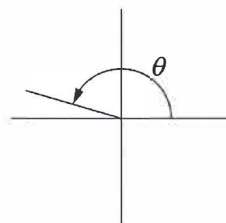
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Which diagram best shows the angle θ ?

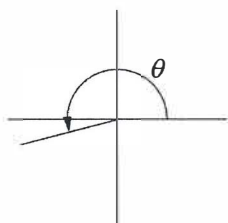
(A)



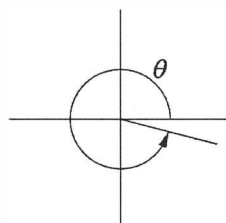
(B)



(C)



(D)



- (c) Solve $\cos \theta = 0.63$ for $-180^\circ \leq \theta \leq 0^\circ$, giving your answer to the nearest degree.

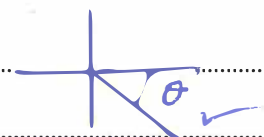
1

$$\cos \theta = 0.63$$

$$\text{ref } \angle = \cos^{-1}(0.63)$$

$$= 51^\circ$$

$$\theta = -51^\circ \quad \checkmark$$



Question 1 Continued

- (d) Find the derivative of $y = \frac{3x^4 - 1}{x}$.

1

$$\begin{aligned} y &= 3x^3 - x^{-1} \\ \frac{dy}{dx} &= 9x^2 - (-1 \times x^{-2}) \\ &= 9x^2 + \frac{1}{x^2} \end{aligned}$$

- (e) Given that $f(x) = x^2 + 1$ and $g(x) = \sqrt{x-3}$. Find $y = f(g(x))$ and state its domain.

2

$$\begin{aligned} f(g(x)) &= (\sqrt{x-3})^2 + 1 \\ &= x-3 + 1 \\ &= x-2 \end{aligned}$$

$$\text{domain of } g(x) = x \geq 3$$

$$\therefore \text{domain of } f(g(x)) = x \geq 3$$

End of Question 1

Question 2 (6 marks)

Marks

- (a) Circle your correct alternative.

1

What is the exact value of $\cos 135^\circ + \operatorname{cosec} 60^\circ$?

(A) $\frac{2\sqrt{2} - \sqrt{3}}{\sqrt{6}}$

(B) $\frac{2\sqrt{2} - 1}{\sqrt{2}}$

(C) $\frac{2\sqrt{2} + \sqrt{3}}{\sqrt{6}}$

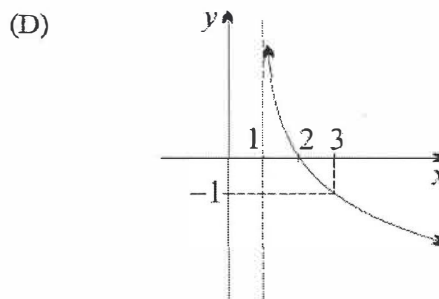
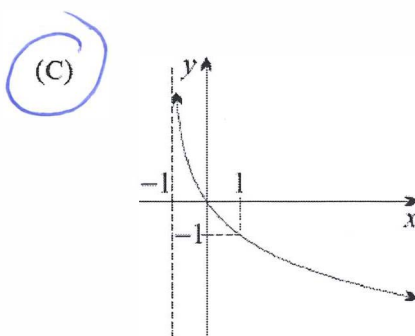
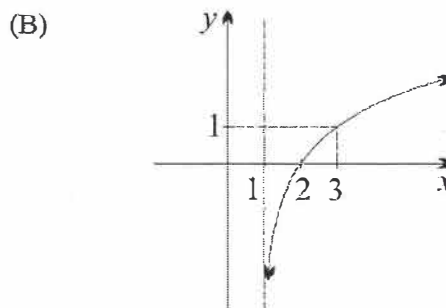
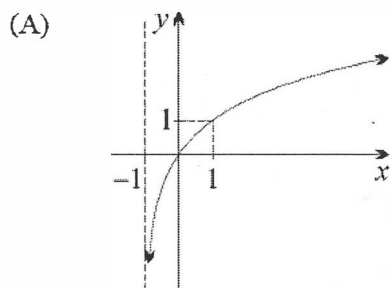
(D) $\frac{2\sqrt{2} + 1}{\sqrt{2}}$

$$\begin{aligned} \cos 135^\circ + \operatorname{cosec} 60^\circ &= -\cos 45^\circ + \frac{1}{\sin 60^\circ} \\ &= -\frac{1}{\sqrt{2}} + \frac{2}{\sqrt{3}} \\ &= \frac{-\sqrt{3} + 2\sqrt{2}}{\sqrt{6}} \end{aligned}$$

- (b) Circle your correct alternative.

1

Which of the following is the graph of $y = -\log_2(x+1)$?



Question 2 Continued

- (c) Solve the equation $|2x - 3| = 6$.

2

$$2x - 3 = 6$$

$$\text{or } 2x - 3 = -6$$

$$2x = 9$$

$$2x = -3$$

$$x = 9/2$$

$$\text{or } x = -3/2$$



- (d) Differentiate with respect to x . Leave your answer in fully factorised form.

2

$$y = \frac{2x^2}{2-x}$$

$$u = 2x^2$$

$$v = 2-x$$

$$u' = 4x$$

$$v' = -1$$

$$y' = \frac{v u' - u v'}{v^2}$$

$$= \frac{(2-x) \times 4x - 2x^2 (-1)}{(2-x)^2}$$



$$= \frac{8x - 4x^2 + 2x^2}{(2-x)^2}$$

$$= \frac{8x - 2x^2}{(2-x)^2}$$

$$= \frac{2x(4-x)}{(2-x)^2}$$



End of Question 2

Question 3 (4 marks)

Marks

- (a) Solve $\sin\left(\frac{x}{2}\right) = \frac{1}{2}$ for $0^\circ \leq x \leq 360^\circ$

2

$$\sin\left(\frac{x}{2}\right) = \frac{1}{2}$$



$$\text{ref } \angle = 30^\circ$$

$$0 \leq x \leq 360^\circ$$

$$\frac{x}{2} = 30^\circ, 150^\circ \text{ (1)}$$

$$0 \leq \frac{x}{2} \leq 180^\circ$$

$$x = 60^\circ, 300^\circ \checkmark \checkmark$$

- (b) Show that $\cot A + \frac{\sin A}{1 + \cos A} = \operatorname{cosec} A$.

2

$$\text{LHS} = \cot A + \frac{\sin A}{1 + \cos A}$$

or

$$= \frac{\cos A}{\sin A} + \frac{\sin A}{1 + \cos A}$$

$$= \frac{\cos A}{\sin A} + \frac{\sin A \times (1 - \cos A)}{1 + \cos A \times (1 - \cos A)}$$

$$= \frac{\cos A(1 + \cos A) + \sin^2 A}{\sin A(1 + \cos A)} \text{ (1)}$$

$$= \frac{\cos A}{\sin A} + \frac{\sin A(1 - \cos A)}{1 - \cos^2 A} \text{ (1)}$$

$$= \frac{\cos A + \cos^2 A + \sin^2 A}{\sin A(1 + \cos A)}$$

$$= \frac{\cos A + \sin A(1 - \cos A)}{\sin^2 A}$$

$$= \frac{\cancel{1} + \cos A}{\sin A(\cancel{1} + \cos A)}$$

$$= \frac{\cos A + 1 - \cos A}{\sin A}$$

$$= \operatorname{cosec} A$$

$$= \frac{1}{\sin A}$$

$$= \text{RHS}$$

$$= \operatorname{cosec} A \checkmark$$

$$= \text{RHS}$$

End of Question 3

Question 4 (4 marks)

Marks

- (a) i) Show that the derivative of $f(x) = x^2 + 3x + 5$ is $f'(x) = 2x + 3$ by first principles.

2

$$f(x) = x^2 + 3x + 5$$

$$f(x+h) = (x+h)^2 + 3(x+h) + 5$$

$$= x^2 + 2xh + h^2 + 3x + 3h + 5$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

must have on every line

$$= \lim_{h \rightarrow 0} \frac{(x^2 + 2xh + h^2 + 3x + 3h + 5) - (x^2 + 3x + 5)}{h} \quad (1)$$

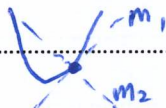
$$= \lim_{h \rightarrow 0} \frac{h(2x + h + 3)}{h}$$

$$= 2x + 3 \text{ as required}$$

must show 'h' cancels for 2 marks

- ii) Hence, find the equation of the normal to the curve $f(x) = x^2 + 3x + 5$ at the point $(1, 9)$.

2



$$f(x) = (2x + 3)$$

$$f(1) = 5$$

$$m_1 m_2 = -1$$

$$m_2 = -\frac{1}{5} \quad (1)$$

$$y - y_1 = m(x - x_1)$$

$$y - 9 = -\frac{1}{5}(x - 1)$$

$$y - 9 = -\frac{1}{5}x + \frac{1}{5}$$

$$y = -\frac{1}{5}x + 9 + \frac{1}{5}$$

$$y = -\frac{1}{5}x + \frac{46}{5}$$

$$5y = -x + 46$$

$$5y + x - 46 = 0$$

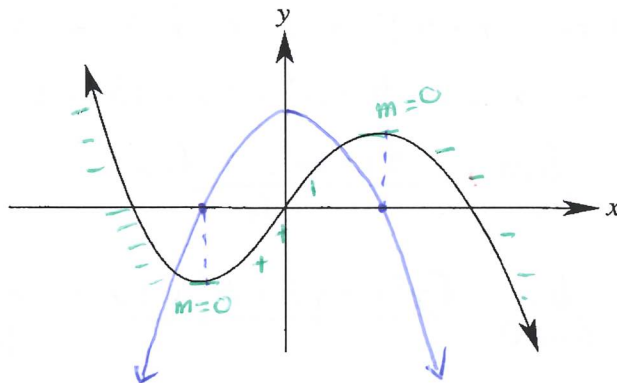
End of Question 4

Question 5 (4 marks)

Marks

- (a) The diagram below shows a sketch of the function $y = f(x)$.
On the same diagram sketch the gradient function $y = f'(x)$.

1



- (b) Show that, $\frac{d}{dx} [3x^2(4x-1)^3] = 6x(10x-1)(4x-1)^2$.

3

$$y = 3x^2(4x-1)^3$$

$$u = 3x^2 \quad v = (4x-1)^3$$

$$u' = 6x \quad v' = 3(4x-1)^2 \times 4 = 12(4x-1)^2$$

①

$$\frac{d}{dx} [3x^2(4x-1)^3] = uv' + vu'$$

$$= 3x^2 \times 12(4x-1)^2 + (4x-1)^3 \times 6x \rightarrow \textcircled{1}$$

$$\Rightarrow 6x(4x-1)^2(6x + 4x-1)$$

$$= 6x(4x-1)^2(10x-1)$$

✓✓✓

factorising should be there for 3rd mark.

End of Question 5

Question 6 (4 marks)

Marks

- (a) Differentiate with respect to x .

$$y = \sqrt{x^2 - 2x}$$

2

$$y = \sqrt{x^2 - 2x}$$

$$y = (x^2 - 2x)^{1/2}$$

$$\frac{dy}{dx} = \frac{1}{2} (x^2 - 2x)^{-1/2} (2x - 2) \quad (1)$$

$$= \frac{1}{2 \sqrt{x^2 - 2x}} \cdot x^2 (x - 1)$$

$$= \frac{x - 1}{\sqrt{x^2 - 2x}}$$

✓✓

or

$$(x - 1) \cdot (x^2 - 2x)^{-1/2}$$

- (b) Solve $(\log_2 x)^2 - 8 = \log_2 x^2$.

2

$$(\log_2 x)^2 - 8 = \log_2 x^2$$

$$(\log_2 x)^2 - 8 - 2 \log_2 x = 0$$

$$u = \log_2 x$$

$$u^2 - 2u - 8 = 0$$

$$(u - 4)(u + 2) = 0$$

(1)

$$\therefore \log_2 x = 4 \quad \text{or} \quad \log_2 x = -2$$

$$x = 2^4$$

as

$$x = 2^{-2}$$

$$x = 2^4$$

or

$$x = \frac{1}{4}$$

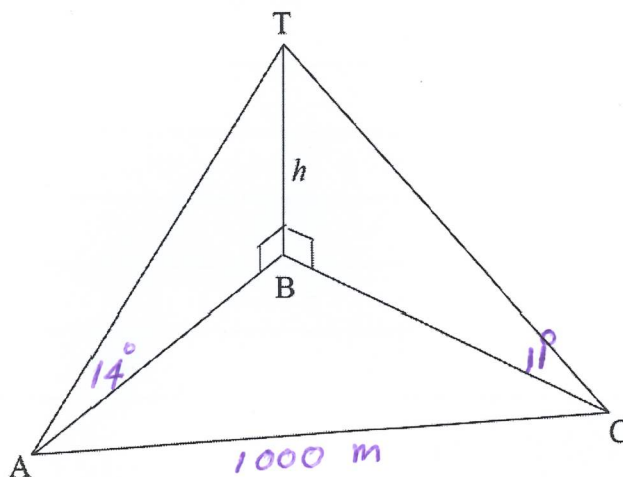
✓✓

End of Question 6

Question 7 (4 marks)

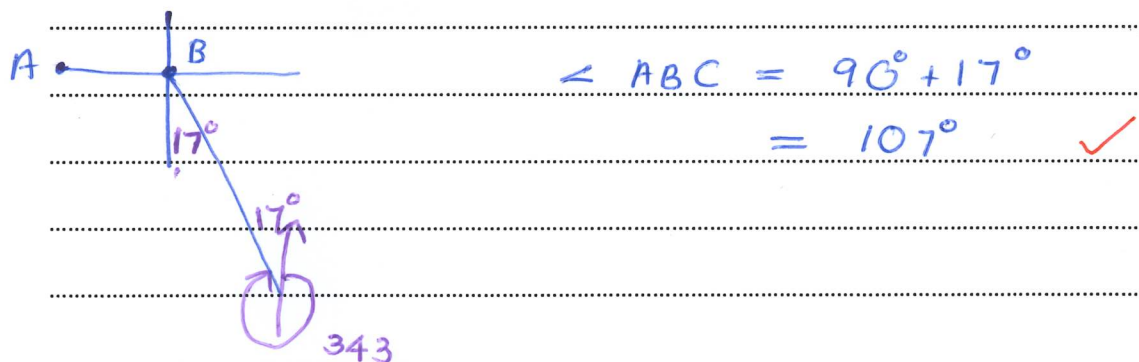
Marks

The angle of elevation of a vertical tower BT of height h metres, at a point A due west of it, is 14° . From the point C, the bearing of the tower is 343° and the angle of elevation is 11° . The points A and C are 1000m apart and on the same level as the base B of the tower.



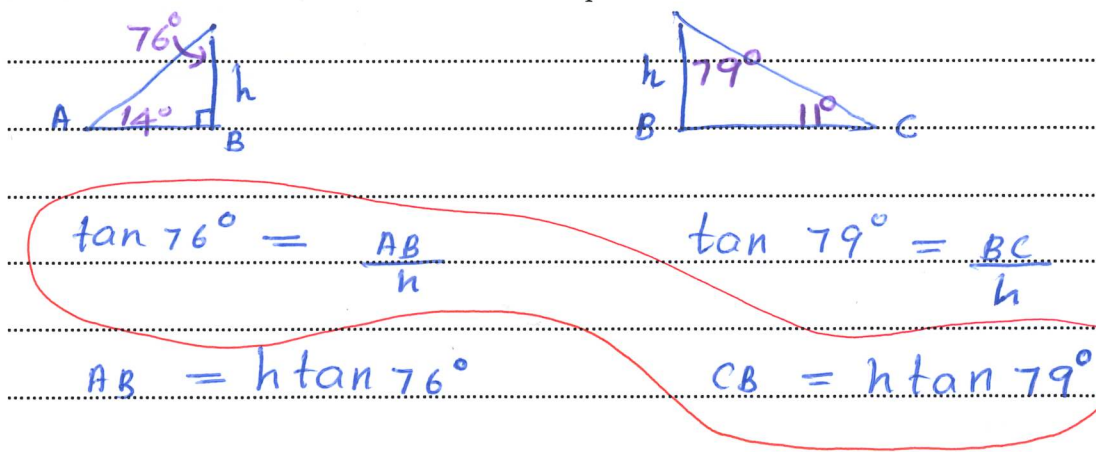
- (a) By drawing a diagram showing the bearings, explain why $\angle ABC = 107^\circ$.

1



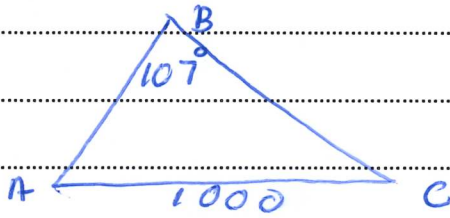
- (b) Show that $AB = h \tan 76^\circ$ and find a similar expression for CB .

1



- (c) Hence, find the height of the tower, correct to the nearest metre.

2



$$\begin{aligned} 1000^2 &= AB^2 + BC^2 - 2 AB \times BC \times \cos 107^\circ \\ &= h^2 \tan^2 76^\circ + h^2 \tan^2 79^\circ - 2 h^2 \tan 76^\circ \tan 79^\circ \cos 107^\circ \\ &= h^2 (\tan^2 76^\circ + \tan^2 79^\circ - 2 \tan 76^\circ \tan 79^\circ \cos 107^\circ) \end{aligned}$$

$$h^2 = \frac{1000^2}{\tan^2 76^\circ + \tan^2 79^\circ - 2 \times \tan 76^\circ \tan 79^\circ \cos 107^\circ}$$

$$h = 135.310$$

$$h \approx 135 \text{ m} \quad \checkmark \quad \checkmark$$

End of Question 7

End of Assessment Task

